

AWS DataSync – Copy Data from an EC2 SMB Share to Amazon S3

To Begin with the Lab

Summary of the Lab

In this lab, **AWS DataSync** was demonstrated in a fully cloud-based setup by using an **Amazon EC2** instance as a simulated on-premises SMB server. The EC2 instance was launched, configured as a shared storage server, and given files to transfer. A DataSync agent was then created using the **Amazon EC2** option, activated, and connected to the shared folder through SMB. After that, a DataSync task was created to copy data from the SMB share to an **Amazon S3** bucket, with the IAM role generated automatically. The transfer was started and verified by checking the task history and confirming that the files appeared in the S3 bucket.

Understanding AWS DataSync for Cloud File Transfers

AWS DataSync is an online data transfer service that automates and speeds up moving data between storage systems such as **Amazon S3**, **Amazon EFS**, **Amazon FSx**, and SMB or NFS file servers. It exists so you do not have to copy files manually when moving data between storage locations. In this lab, the source was an SMB share running on an **Amazon EC2** instance, and the destination was **Amazon S3**. The benefit is secure, fast, and repeatable file movement with very little manual work.

Example:

A Windows **EC2** server named Storage Server stores 11 files in a shared folder named data. The team wants to move those files to a bucket named cloud storage 02 without using manual upload and download steps. DataSync reads the SMB share and copies the files into S3, then the same files appear in the bucket after the transfer completes.

Lab Steps

Step 1 – Create the storage server EC2 instance

- Sign in to the **AWS Management Console**.
- Open **Amazon EC2**.

Resources

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)	1	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	2	Instances	2
Key pairs	31	Load balancers	0	Placement groups	0
Security groups	39	Snapshots	0	Volumes	2

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

[Launch instance](#) [Migrate a server](#)

Note: Your instances will launch in the United States (N. Virginia) Region

Instance alarms

0 in alarm 0 OK 0 insufficient data

[View in CloudWatch](#)

Scheduled events

United States (N. Virginia)

Service health

[AWS Health Dashboard](#)

Zones

Zone name	Zone ID
us-east-1a	use1-az4
us-east-1b	use1-az6
us-east-1c	use1-az1
us-east-1d	use1-az2
us-east-1e	use1-az3
us-east-1f	use1-az5
us-east-1-atl-2a	use1-atl2-az1

[Enable additional Zones](#)

EC2 cost

Date range: Past 6 months Region: Global

Total cost

Regions

[Analyze your costs in Cost Explorer](#)

Account attributes

Default VPC [vpc-099792461c7d0a616](#)

Settings

Data protection and security

Allowed AMIs

Zones

EC2 Serial Console

Default credit specification

Managed resources

EC2 console preferences

Explore AWS

- Launch a new instance named **Storage Server**.
- Use a **Windows Server 2019** AMI for the shared storage server.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents My AMIs Quick Start

Amazon Linux macOS Ubuntu **Windows** Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Microsoft Windows Server 2019 Base [Free tier eligible](#)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Microsoft Windows 2019 Datacenter edition, [English]

Summary

Number of instances: 1

Software Image (AMI): Microsoft Windows Server 2019 ...[read more](#)

Virtual server type (instance type): t3.small

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 30 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

[Cancel](#) [Launch instance](#) [Preview code](#)

- Select **t3.small** for the instance type for lab purpose.
- Keep the security group open for all traffic during the lab.

Instance type
t3.small

Key pair (login)
Key pair name - required: June-2026-us

Network settings
Network: vpc-099792461c7d0a616
Subnet: No preference (Default subnet in any availability zone)
Auto-assign public IP: Enable
Firewall (security groups): Create security group

Summary
Number of instances: 1
Software Image (AMI): Microsoft Windows Server 2019 ...
Virtual server type (instance type): t3.small
Firewall (security group): New security group
Storage (volumes): 1 volume(s) - 30 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

- Create and add **Allow All Traffic** rule as well.

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type: ssh
Protocol: TCP
Port range: 22
Source type: Anywhere
Source: 0.0.0.0/0
Description - optional: e.g. SSH for admin desktop

Security group rule 2 (All, All, 0.0.0.0/0)

Type: All traffic
Protocol: All
Port range: All
Source type: Anywhere
Source: 0.0.0.0/0
Description - optional: e.g. SSH for admin desktop

Summary
Number of instances: 1
Software Image (AMI): aws-datasync-2.0.1784035868.1...
Virtual server type (instance type): m7i-flex.large
Firewall (security group): New security group
Storage (volumes): 1 volume(s) - 80 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

- Launch the instance and wait for it to become available.

Step 2 – Configure the Windows shared folder

- Connect to the EC2 instance through RDP.
- Download the remote desktop file and open it.
- Click on get password and paste your .pem file.

EC2 > Instances > i-04153be8577c5862a > Connect to instance

Instance ID: i-04153be8577c5862a (Storage Server)

VPC ID: vpc-099792461c7d0a616

Security groups: sg-0e0e25756447c708a (launch-wizard-20)

IAM role: -

SSM Session Manager **RDP client** EC2 serial console

Instance ID: i-04153be8577c5862a (Storage Server)

Connection Type: **Connect using RDP client**

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

Download remote desktop file

When prompted, connect to your instance using the following username and password:

Public DNS: ec2-54-145-62-1.compute-1.amazonaws.com

Username: Administrator

Password: **Get password**

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

Cancel

- Click on Decrypt Password and paste it.

EC2 > Instances > i-04153be8577c5862a > Get Windows password

Get Windows password info

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID: i-04153be8577c5862a (Storage Server)

Key pair associated with this instance: June-2026-us

Private key: Either upload your private key file or copy and paste its contents into the field below.

Upload private key file

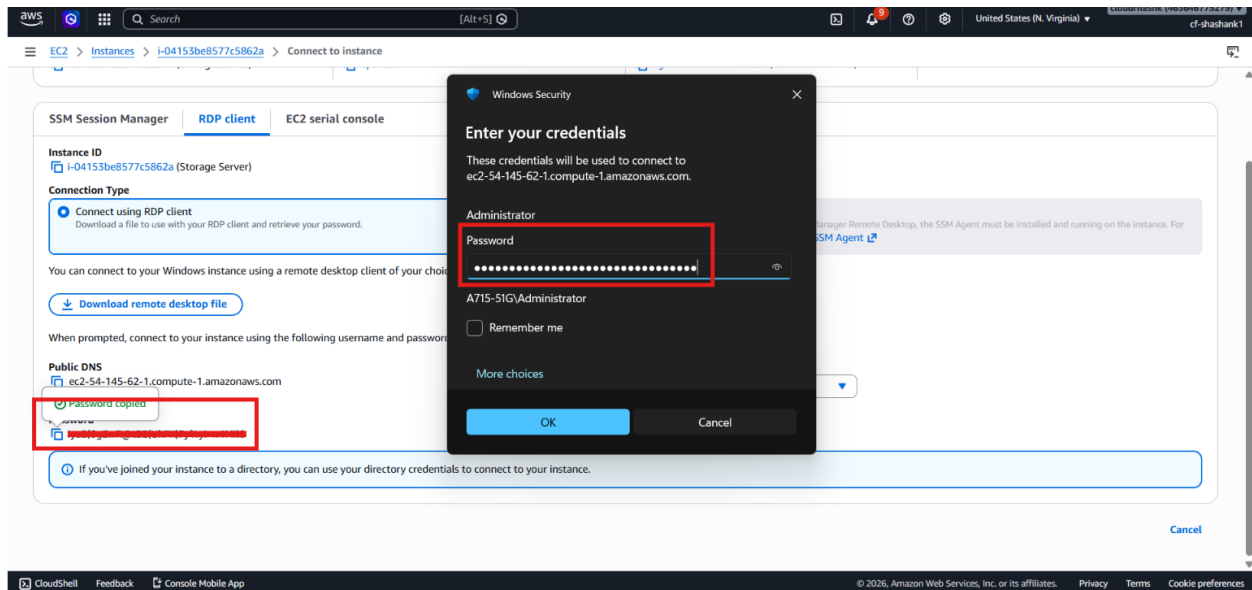
June-2026-us.pem
1.68 KB

Private key contents

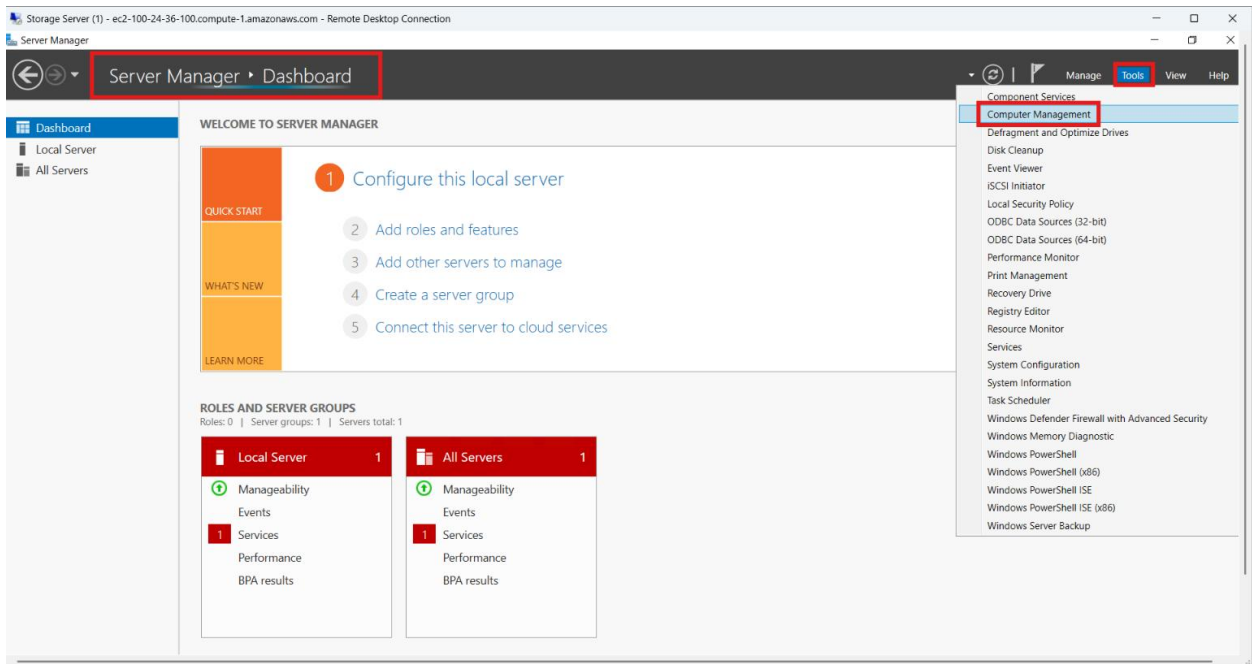
-----BEGIN RSA PRIVATE KEY-----
[Redacted private key content]

Cancel **Decrypt password**

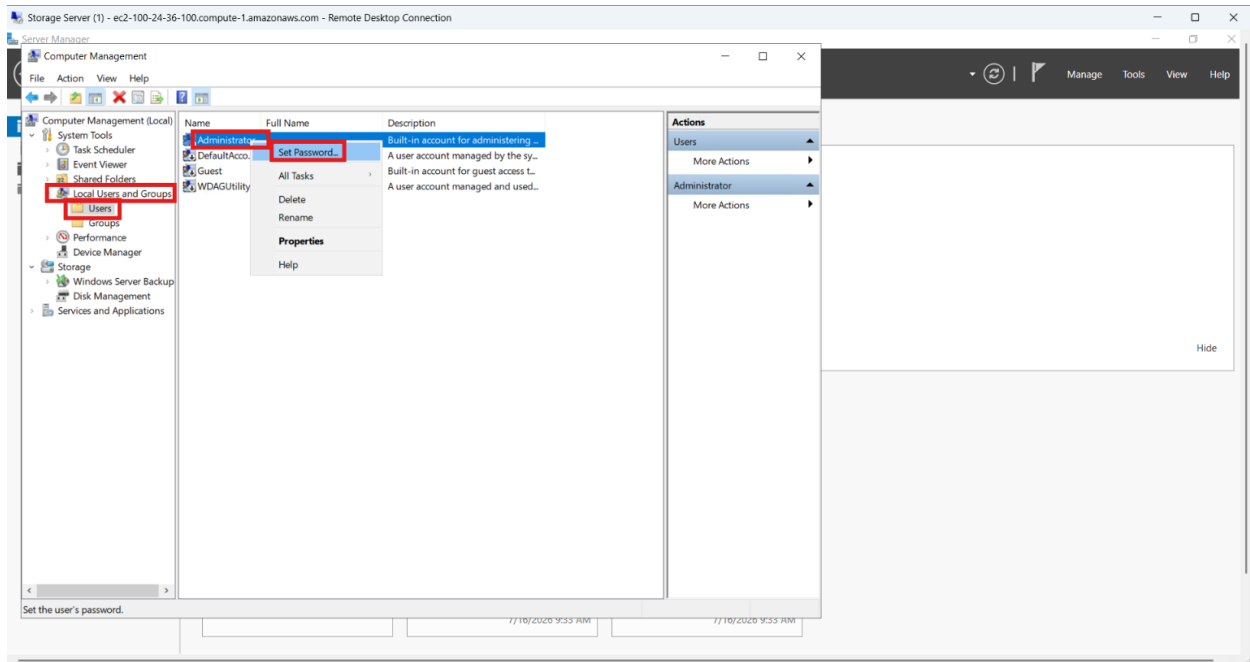
- Paste the password in the panel to connect to the Server.
- Log in as the administrator user.



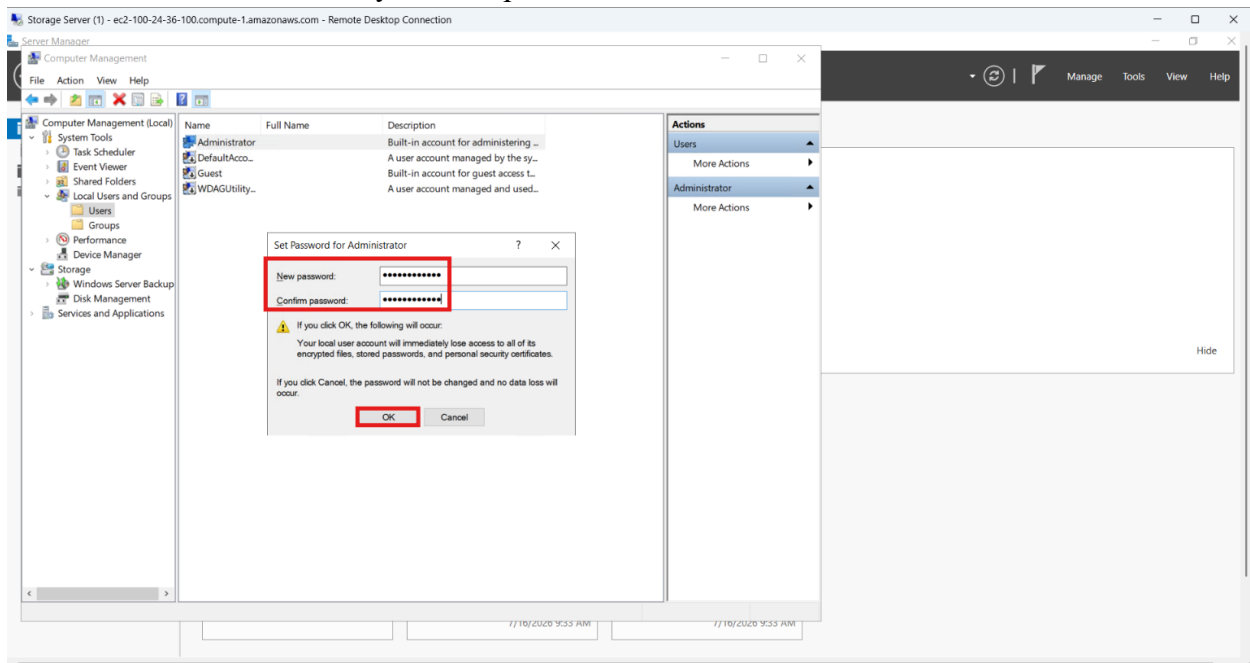
- Change the administrator password, for that Open Server Manager → Tools and Open Computer Management.



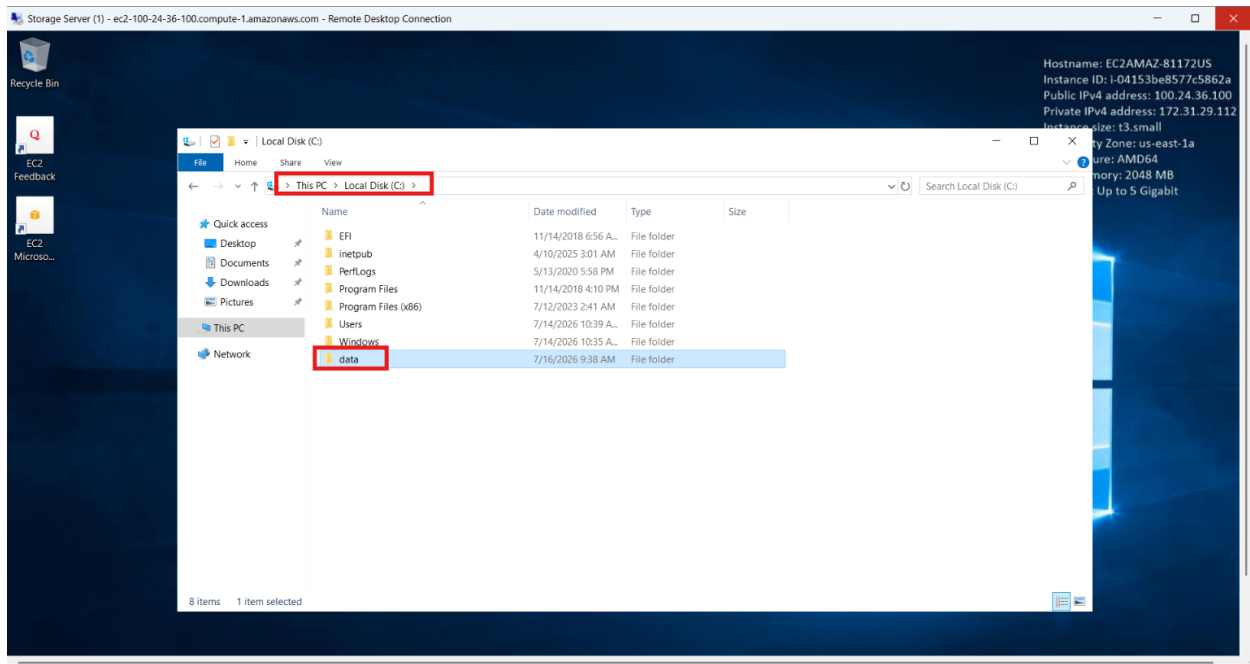
- Now Go to **Local Users and Groups** → **Users** → Right-click on **Administrator** → Click on **Set Password**.



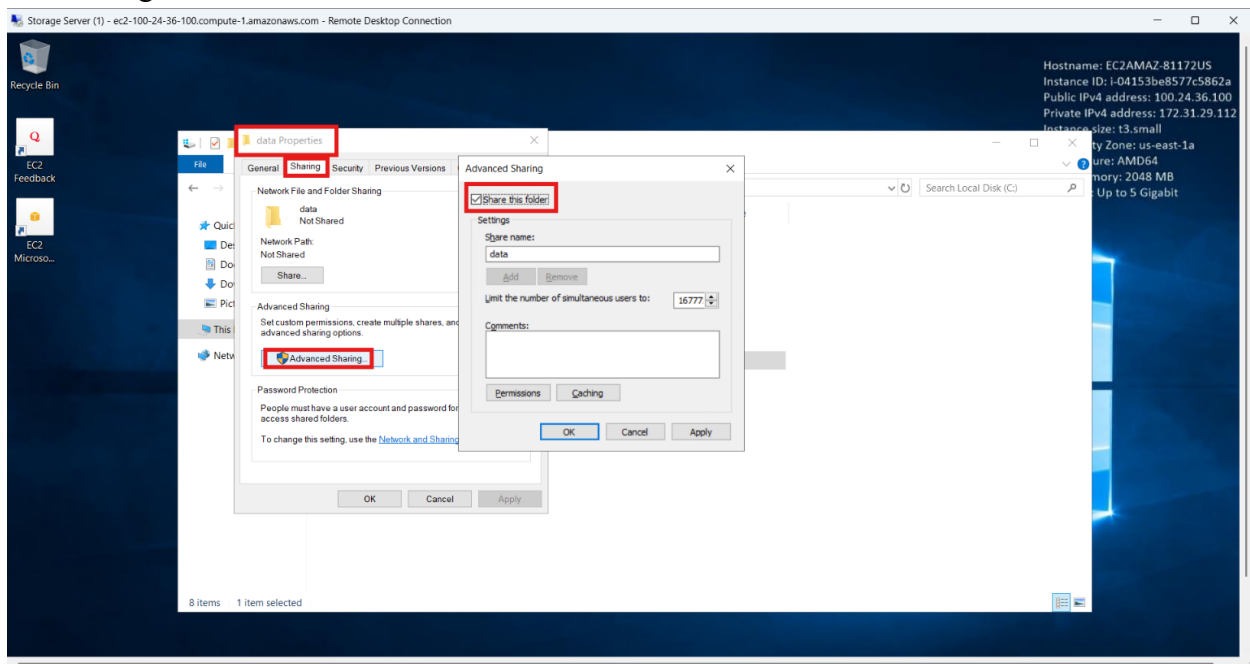
- Click **Proceed** and enter your new password and click **OK**.



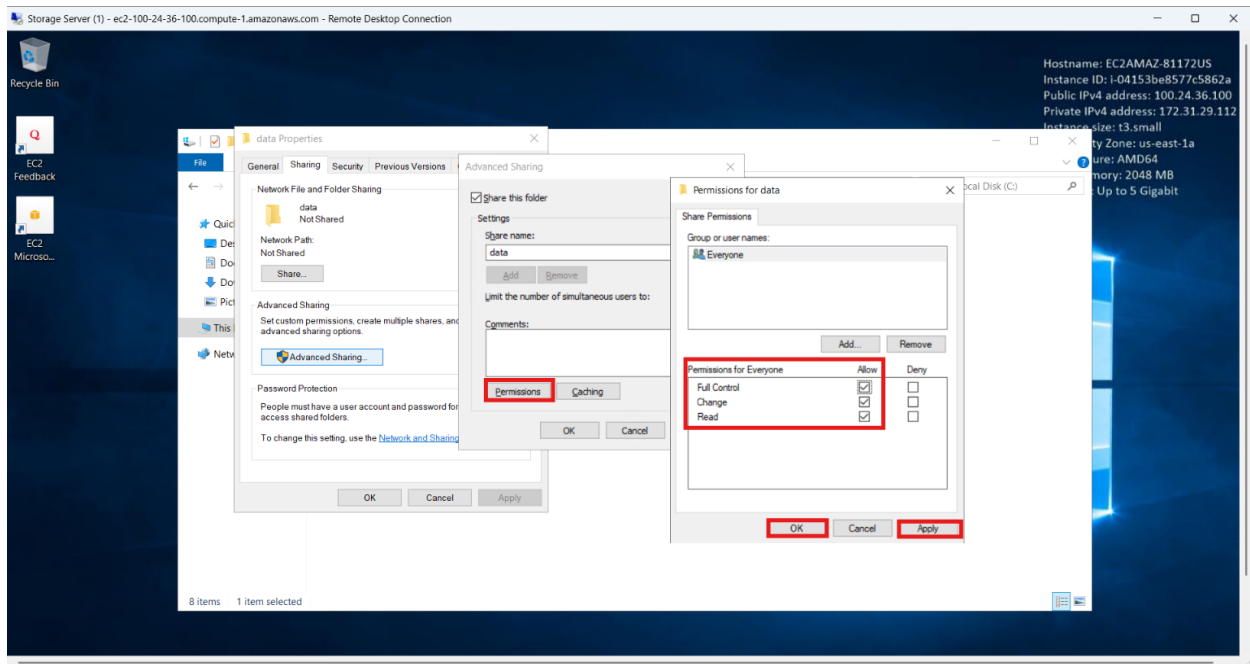
- Now open **C: Drive** and create a shared folder named **data**.



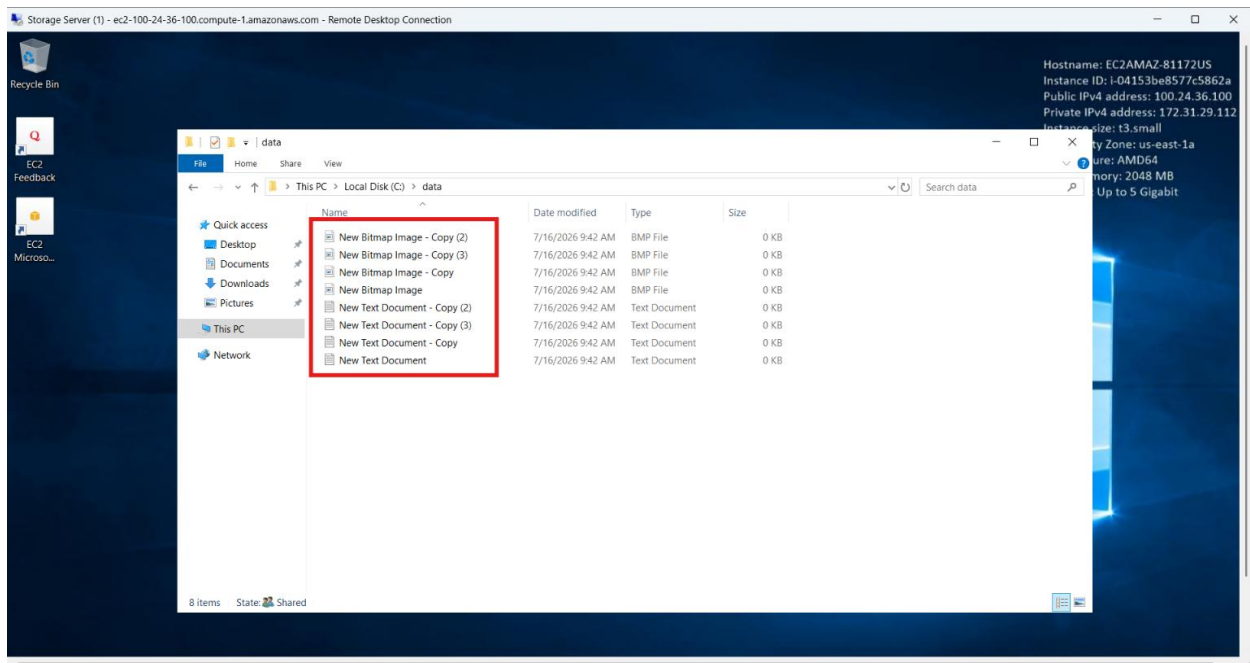
- Open the folder **Properties** by Right-clicking on it and go to **sharing** tab → **Advanced Sharing** and click on **Share this Folder**.



- Click on **Permissions** and give everyone **full control permissions** and click **Apply** and **OK**.

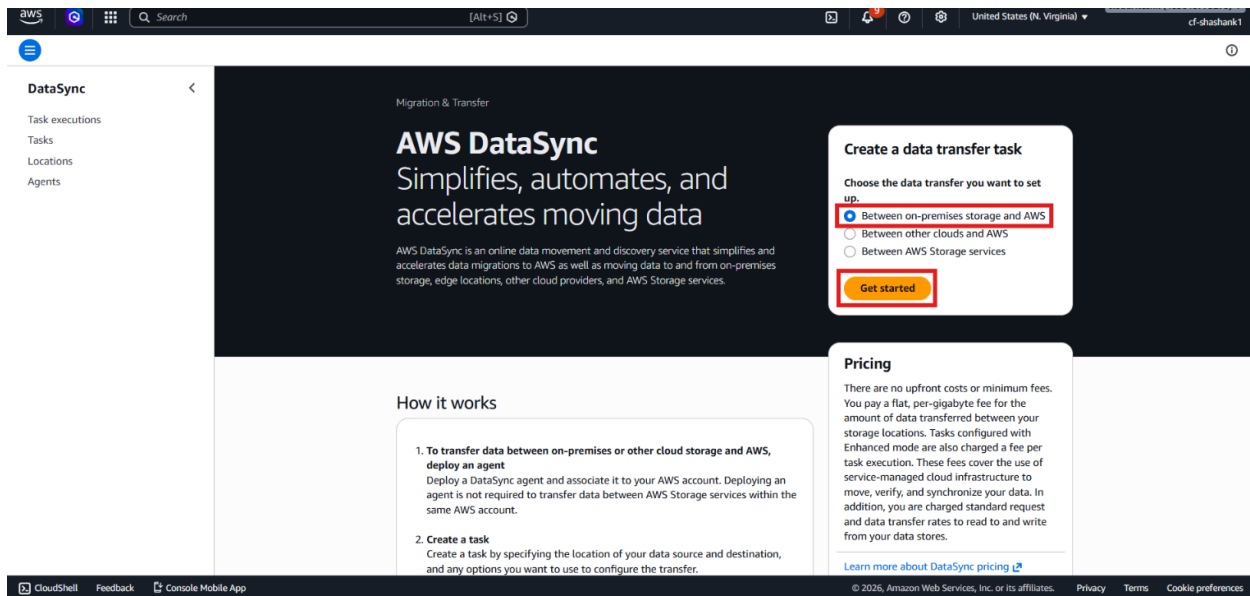


- Add a few test files into the shared folder.

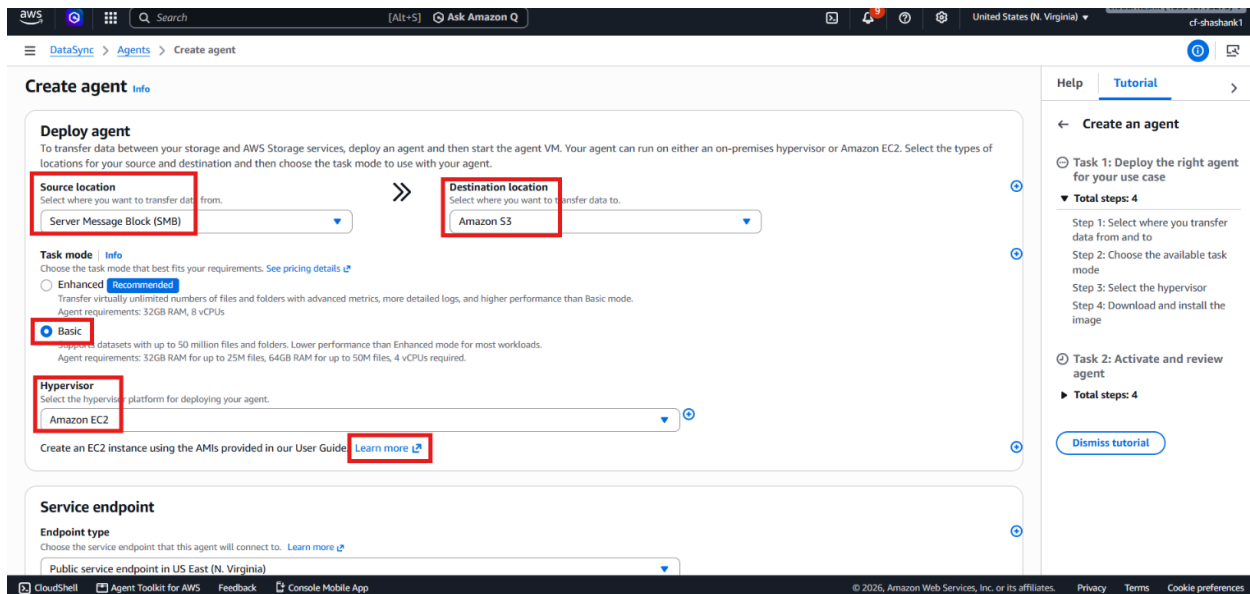


Step 3 – Create and activate the DataSync agent

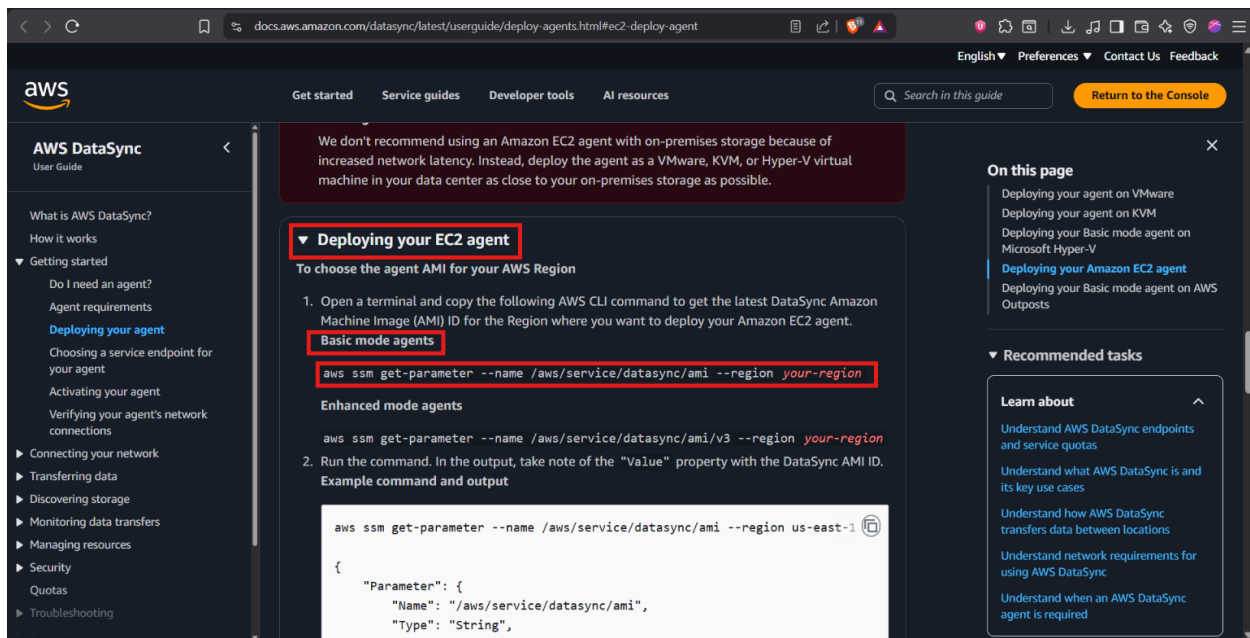
- Open **AWS DataSync**.
- Choose the option for transferring data between **on-premises storage** and **AWS**.



- Choose Source location as **Server Message Block (SMB)** and Destination as **Amazon S3**.
- Choose it as **Basic** Task Mode.
- Select **Amazon EC2** as the hypervisor type.
- Now click on **Learn More**.



- Use the AWS official AMI details for the DataSync agent.



- Now configure AWS on your local system as we did in earlier lab by running “aws configure” and then run the AMI command from the official site.

```

Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Shashank>aws configure
AWS Access Key ID [*****KBXE]:
AWS Secret Access Key [*****nahs]:
Default region name [us-east-1]:
Default output format [json]:

C:\Users\Shashank>aws ssm get-parameter --name /aws/service/datasync/ami --region us-east-1
{
  "Parameter": {
    "Name": "/aws/service/datasync/ami",
    "Type": "String",
    "Value": "ami-06643caf2bfff83e9a",
    "Version": 183,
    "LastModifiedDate": "2026-07-15T18:58:00.914000+05:30",
    "ARN": "arn:aws:ssm:us-east-1:parameter/aws/service/datasync/ami",
    "DataType": "text"
  }
}

C:\Users\Shashank>

```

- Now copy this AMI ID and keep it somewhere safe as we are going to make our DataSync Agent with this AMI ID.
- Now visit the site and copy the following command and replace the required values.

The screenshot shows the AWS DataSync User Guide page for deploying an Amazon EC2 agent. The main content area includes a tip about avoiding charges for transferring across Availability Zones and a list of steps to follow. The first step is to copy the following URL:

```
https://console.aws.amazon.com/ec2/v2/home?region=agent-region#LaunchInst
```

The URL contains two placeholders: *agent-region* and *ami-id*. The instructions specify:

- Replace *agent-region* with the Region where you want to deploy your agent.
- Replace *ami-id* with the DataSync AMI ID that you obtained.

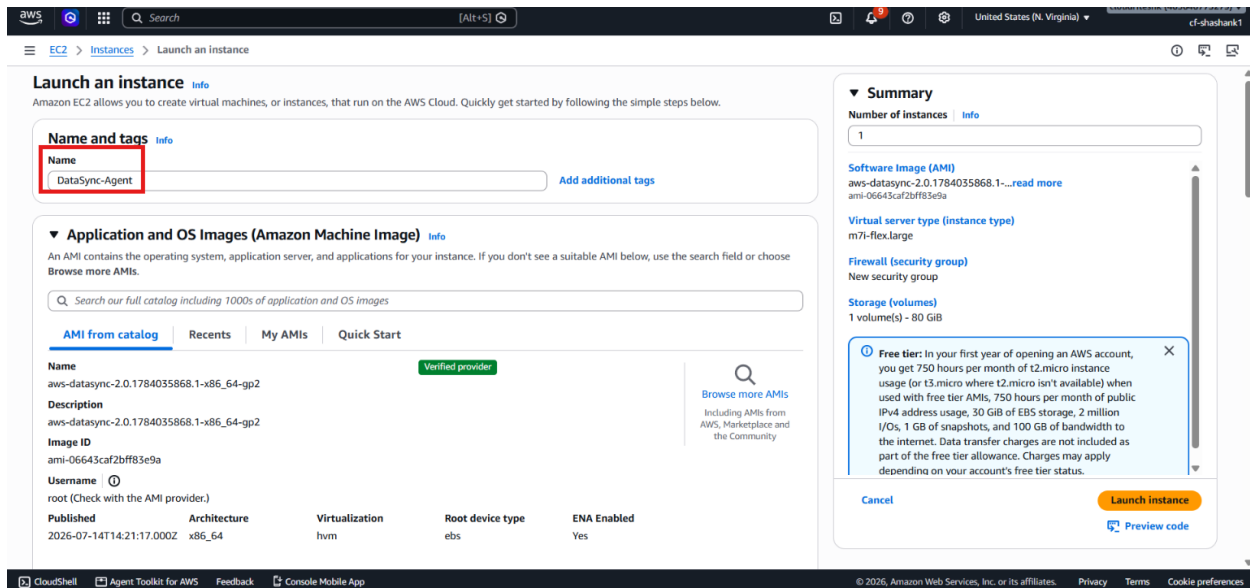
The subsequent steps are:

- Paste the URL into a browser.
- The Amazon EC2 instance launch page in the AWS Management Console displays.
- For **Instance type**, choose one of the recommended Amazon EC2 instances for DataSync.
- For **Key pair**, choose an existing key pair, or create a new one.
- For **Network settings**, choose **Edit** and then do the following:

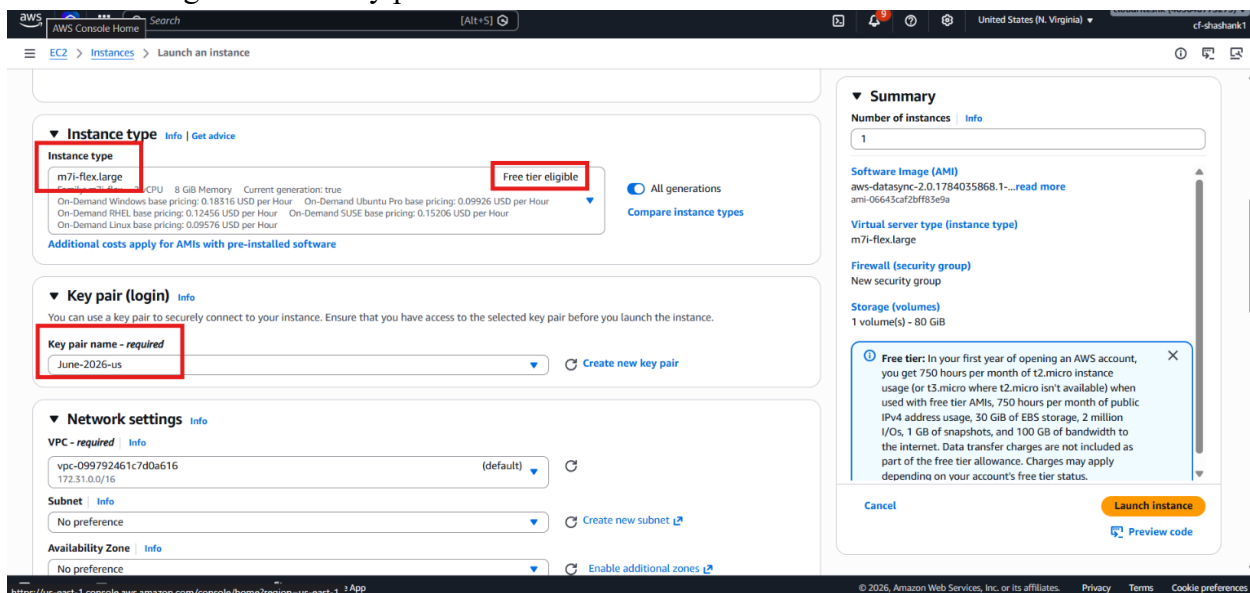
- And paste the modified URL to the browser and press enter.

The screenshot shows a web browser with the AWS console URL in the address bar. The URL is `https://console.aws.amazon.com/ec2/v2/home?region=us-east-1#LaunchInstanceWizard:ami=ami-06643caf2bf83e9a`. The browser also displays a large clock showing 1:28 and the text "Never give up!".

- Enter the name: DataSync-Agent.



- For the agent, we must provide minimum of 8GB RAM so we must change the instance and then configure it with key pair.



- Now configure the Network settings and allow **All Traffic** from **Anywhere**.
- Launch the agent instance.

Launch an instance

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./@.#[]+=&()!\$*

Description - required [Info](#)

launch-wizard-22 created 2026-07-16T08:00:21.077Z

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type [Info](#) Protocol [Info](#) Port range [Info](#)

ssh TCP 22

Source type [Info](#) Source [Info](#) Description - optional [Info](#)

Anywhere Add CIDR, prefix list or security group e.g. SSH for admin desktop

0.0.0.0/0 X

Security group rule 2 (All, All, 0.0.0.0/0)

Type [Info](#) Protocol [Info](#) Port range [Info](#)

All traffic All All

Source type [Info](#) Source [Info](#) Description - optional [Info](#)

Anywhere Add CIDR, prefix list or security group e.g. SSH for admin desktop

0.0.0.0/0 X

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Summary

Number of instances [Info](#)

1

Software Image (AMI)

aws-datasync-2.0.1784035868.1...[read more](#)

ami-06645ca72bf85e9a

Virtual server type (instance type)

m7i-flex.large

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 80 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

Cancel Launch instance Preview code

- Activate the agent by entering its Public IPv4 address in DataSync and press **Get Key**.

Create agent

Amazon EC2

Create an EC2 instance using the AMIs provided in our User Guide. [Learn more](#)

Service endpoint

Endpoint type

Choose the service endpoint that this agent will connect to. [Learn more](#)

Public service endpoint in US East (N. Virginia)

Activation key

Activation securely associates the agent that you deployed with your AWS account. [Learn more](#)

Choose a method for obtaining your agent's activation key. [Learn more](#)

Automatically get the activation key from your agent

Your browser will connect to your agent over HTTP to get a unique activation key.

Manually enter your agent's activation key

Choose this option if you already have an activation key and need to enter it manually.

Agent address

When your DataSync agent has been deployed, enter its domain name or IP address. When you click **Get key**, your browser will connect to this address to get a unique activation key.

http://18.206.189.34

Make sure this address is accessible from your browser.

Cancel Get key

- The IP Address can be found by clicking on the Instance we have created.

The screenshot shows the AWS Management Console interface. On the left, the navigation menu includes 'EC2', 'Instances', 'Images', 'Elastic Block Store', and 'Network & Security'. The main content area displays a list of EC2 instances. The 'DataSync-Agent' instance is selected. Below the list, the 'Details' tab for instance 'i-062a697a11ffcd5aa' is open. A red box highlights the 'Public IPv4 address copied' status in the 'Networking' section.

- The agent will be activated, Give name to your agent and click **Create Agent**.

The screenshot shows the AWS DataSync 'Create agent' page. The 'Activation key' section shows a successful retrieval of the activation key. The 'Agent name' field is highlighted with a red box and contains the text 'Ag1'. The 'Create agent' button is also highlighted with a red box.

- Create the agent and wait until it becomes active.

Step 4 – Create the S3 destination bucket

- Open **Amazon S3**.
- Create a new bucket named **amazon-datasync-ec2**.
- Confirm that the bucket is empty before the transfer.

Step 5 – Create the DataSync task

- Return to **AWS DataSync**.
- Click on **Create task**.

The screenshot shows the AWS DataSync console. On the left, there's a sidebar with 'DataSync' and its sub-sections: 'Task executions', 'Tasks', 'Locations', and 'Agents'. The main area is divided into two sections: 'Locations (0)' and 'Tasks (0)'. The 'Locations' section has a search bar and a table with columns: Location ID, Host, Path, Type, Created, Tasks, and Agents. Below the table, it says 'No locations. You don't have any locations in us-east-1.' The 'Tasks' section also has a search bar and a table with columns: Task ID, Name, Status, Mode, Schedule, Source, Destination, and Created. Below the table, it says 'No tasks. You don't have any tasks using this agent.' In the top right of the 'Tasks' section, there's a 'Create task' button highlighted with a red box.

- Create a new location for the source.
- Select **SMB** as the source type.
- Enter the **Public IP address** of the shared Storage Server.

The screenshot shows the AWS EC2 console. On the left, there's a sidebar with 'EC2' and its sub-sections: 'Dashboard', 'AWS Global View', 'Events', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', 'Images', 'AMI Catalog', 'Elastic Block Store', 'Volumes', 'Snapshots', 'Lifecycle Manager', 'Network & Security', 'Security Groups', 'Elastic IPs', 'Placement Groups', 'Key Pairs', and 'Network Interfaces'. The main area shows a list of instances. The 'Storage Server' instance is highlighted with a red box. Below the list, the details for the 'Storage Server' instance are shown. The 'Public IPv4 address' is 100.24.36.100, which is highlighted with a red box. Other details include 'Instance state: Running', 'Private IP DNS name (IPv4 only): ip-172-31-29-112.ec2.internal', and 'Instance type: t3.small'.

- Enter the SMB share name **/data**.

Choose the location that you want to transfer data from

☒ Create a new location ☐ Choose an existing location

Configuration

Location type
Server Message Block (SMB)

Region
US East (N. Virginia) us-east-1
Transfers to or from SMB must stay in the current Region.

Agents
The agents you choose should have access to your SMB server.
Choose one or more agents [Create agent](#)

Ag1 (agent-0952570533df0d033a) Public - US East (N. Virginia)
Basic mode

SMB server
Domain name or IP address of the SMB server.
172.31.29.112

Share name
Mount path exported by the SMB server or a subdirectory of that path, delimited with forward slashes (for example, /my-folder/sub-folder).
/data

► Additional settings

- Choose **NTLM** as Authentication Type.
- Use **administrator** as the username.
- Enter the Windows password for the shared server we have configured earlier.

Authentication Info

Authentication type
DataSync authenticates with the SMB server using this method.
☒ NTLM ☐ Kerberos

ⓘ DataSync authenticates using a user name and a password that you provide. [Learn more about choosing a user name](#) that ensures sufficient permissions to files, folders and metadata.

User
User who has permission to access the SMB server.
administrator

Access configuration
☒ Specify a password ☐ Use a Secrets Manager secret

Password
Password of the user who has permission to access the SMB server.

KMS Key Arn (optional)
Enter an optional KMS Key ARN to encrypt your password.
[Choose an AWS KMS key or enter an ARN](#)

Domain - optional
The name of the Windows domain that the SMB server belongs to.

- Select **Amazon S3** as the destination.
- Choose the bucket **amazon-datasync-ec2**.
- Keep the storage class as standard.
- Let DataSync auto-generate the IAM role by clicking on **Autogenerate** button.

Configuration

Location type
Amazon S3

Region
US East (N. Virginia) us-east-1

S3 URI
s3://amazon-datasync-ec2

S3 storage class
Standard

IAM role
Choose a role with access to your S3 bucket

Tags - optional

- Enter the name for the Task.
- Choose the Mode to **Basic**.
- Choose **Everything** for the Contents to Scan option.

Configure settings

Task mode
Basic

Name - optional
Agent-EC2

Source data options
Contents to scan: Everything

Excludes - optional

- Set Transfer mode to **Transfer all data**.
- Leave rest settings as default.

Transfer options

Transfer mode
Determines whether DataSync transfers only the data and metadata that differ between the source and the destination.

Transfer all data

Verification
Specify how and when DataSync checks the integrity of your data. DataSync always performs data integrity checks during a transfer.

Verify only transferred data

Bandwidth limit
Allocate the maximum network bandwidth to be utilized by this task.

☒ Use available

File and object handling

☒ Keep deleted files
Keep or delete data in your destination that isn't in your source. You can't deselect this option if you set Transfer mode to Transfer all data. [Learn more](#)

☒ Overwrite files
Modify or preserve your destination data when your source data (including metadata) changes. You can't deselect this option if you set Transfer mode to Transfer all data. [Learn more](#)

Additional settings

Schedule
Specify when you want this task to run.

Frequency
Not scheduled

- Create the task.

Step 6 – Start the transfer

- Open the DataSync task we have created.
- Start the task manually with **Start** → **Start with defaults**.

DataSync < **Agent-EC2**

Start Stop Edit Delete

Start with defaults
Start with overriding options

Overview

Task status
Available

Latest execution
-

Task mode
Basic

Task ARN
arn:aws:datasync:us-east-1:123456789012:task/task-0e972524246709dbf

Creation time
Jul 16, 2026, 3:38:03 PM GMT+5:30

Locations Options Schedule Tags Monitoring Task executions

Source location

Location ID
loc-066873a3b3cea38d7

Region
US East (N. Virginia) us-east-1

Type
Server Message Block (SMB)

Path
/data/

Creation Time
Jul 16, 2026, 3:38:02 PM

Server
172.31.29.112

Agents
agent-0952570533df0033a
Basic mode

SMB version
Automatic

Authentication type
NTLM

User
administrator

Secret
aws-datasyncloc-066873a3b3cea38d7-TgMYrC

Destination location

Location ID
loc-0ee01bbb72a4cb25c

Region
US East (N. Virginia) us-east-1

Type
Amazon S3

Path
/

Creation Time
Jul 16, 2026, 3:38:03 PM GMT+5:30

S3 bucket
amazon-datasync-ec2

S3 storage class
Standard

S3 bucket IAM role
AWSDataSyncS3BucketAccess-amazon-datasync-ec2-d3649

- Wait for the transfer to finish.

Step 7 – Verify the transfer

- Open the task execution history.

Agent-EC2

Overview

Task status: Available | Latest execution: Jul 16, 2026, 3:52:15 PM GMT+5:30 | Task mode: Basic | Creation time: Jul 16, 2026, 3:43:07 PM GMT+5:30

Task ARN: arn:aws:datasync:us-east-1:task/task-071169190a392a4c0

Locations | Options | Schedule | Tags | Monitoring | **Task executions**

Task executions (2)

A task execution is an individual run of a DataSync task and contains information about the execution results.

Execution ID	Start time	Duration	Status	Data throughput	File throughput	Mode
exec-0d5ebbcd86f763624	Jul 16, 2026, 3:52:15 PM GMT+5:30	2 seconds	Success	0 B/s	8.26 file/s	Basic
exec-076fb7505ed1812e1	Jul 16, 2026, 3:43:37 PM GMT+5:30	0 seconds	Error	0 B/s	0 file/s	Basic

- Confirm that the run completed successfully.

exec-0d5ebbcd86f763624

Overview

Execution status: Success | Start time: Jul 16, 2026, 3:52:15 PM GMT+5:30 | Task ID: task-071169190a392a4c0 | Duration: 2 seconds | End time: Jul 16, 2026, 3:54:32 PM GMT+5:30 | Task mode: Basic | Task execution ARN: arn:aws:datasync:us-east-1:task/task-071169190a392a4c0/execution/exec-0d5ebbcd86f763624

Performance

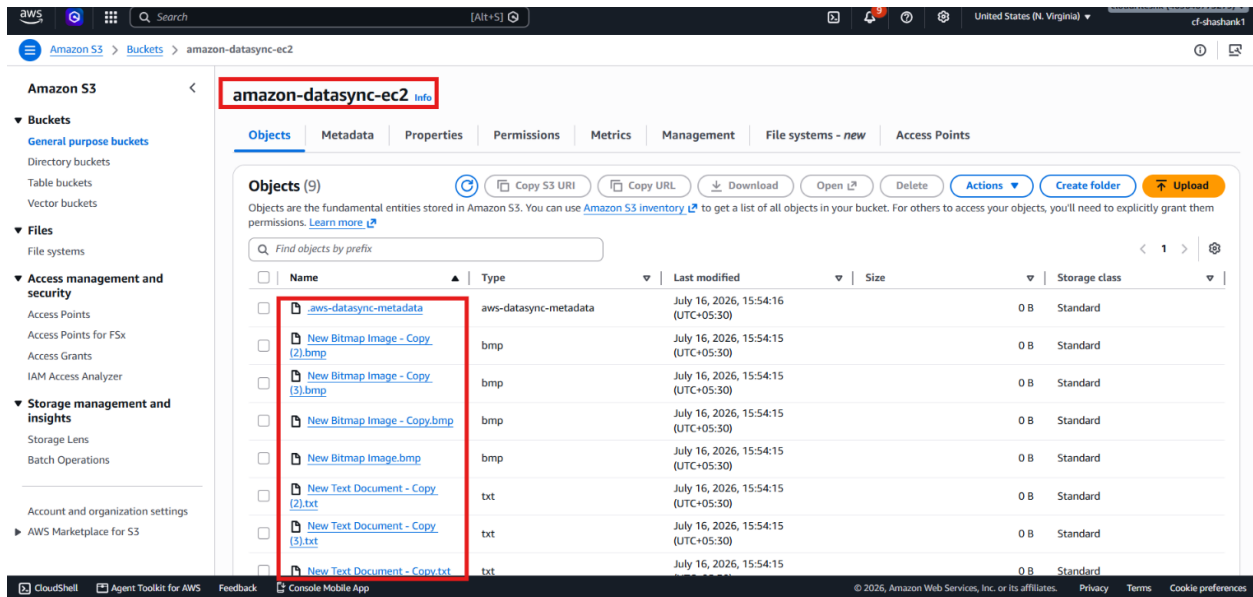
Transferred: 9 files | Skipped: 0 files | Remaining: 0 files | Verified: 9 files | Data transferred: 0 B | Data throughput: 0 B/s | File throughput: 8.26 file/s | Percent compressed: -

Processing rate (file/s): No data available. Try adjusting the dashboard time range.

Data throughput (MiB/s): No data available. Try adjusting the dashboard time range.

Network throughput (Gbps): No data available. Try adjusting the dashboard time range.

- Check the file count in the transfer summary.
- Open the S3 bucket.
- Refresh the bucket contents.
- Confirm that all files now appear in **Amazon S3**.



Verification

- Verified that the Windows **EC2** storage server was created successfully.
- Verified that the shared folder data was created and populated with test files.
- Verified that the **AWS DataSync** agent was created and activated.
- Verified that the SMB source location pointed to the correct server and share.
- Verified that the destination bucket cloud storage 02 was created successfully.
- Verified that the DataSync task completed successfully.
- Verified that the files appeared in **Amazon S3** after the transfer.